**Assessment status: Preliminary.** Reviewed and approved as Preliminary.

# DEEP Metrics

## Northeastern Highlands reference assessment | Preliminary

| | |
|---|---|
| Stream | (unnamed reach) |
| Coordinates | 43.7, -72.2 |
| COMID / HUC8 | 9327042 / 01080104 |
| Drainage area | 3.0 km2 |
| Reach length | 1234.5 ft |
| Watershed basis | NHDPlus V2 basin of the nearest StreamCat reach (StreamCat lookup engine) |
| StreamCat reach | COMID 9327042 (this reach) |

| Function | Metric | F/D | How it is measured | Status | Value | Source |
|---|---|---|---|---|---|---|
| Catchment hydrology | Impervious surface (%) | D | Percent of the delineated watershed in impervious surface, computed by DEEP. No field measurement. | Available | 3.0 | NLCD 2021 |
| Catchment hydrology | Agriculture, crop and hay (%) | D | Percent of the delineated watershed in cultivated crops plus hay and pasture, computed by DEEP. No field measurement. | Available | 12.75 | EPA StreamCat |
| Surface water storage | Wetland (%) | D | Percent of the watershed in woody plus herbaceous wetland (NLCD 2019), read from EPA StreamCat for the reach. No field measurement. | Reference only | 0.4 | Computed by DEEP |
| Reach inflow | Road density (km/sq km) | D | Road length per square kilometer of the delineated watershed, computed by DEEP. No field measurement. | Available | 3.0 | NLCD 2021 |
| Streamflow regime | Base flow index (%) | D | Percent of streamflow that comes from base flow in the watershed (USGS base flow index grid), read from EPA StreamCat for the reach. No field measurement. | Available | 12.75 | EPA StreamCat |
| Streamflow regime | Degree of regulation (%) | D | Normal storage of upstream dams as a percent of mean annual runoff, both read from EPA StreamCat for the reach. No field measurement. | Available | 0.4 | Computed by DEEP |
| Low flow and baseflow dynamics | Base flow index (%) | D | Percent of streamflow that comes from base flow in the watershed (USGS base flow index grid), read from EPA StreamCat for the reach. No field measurement. | Available | 12.75 | EPA StreamCat |
| Low flow and baseflow dynamics | Bankfull width:depth ratio | F | At each transect, measure bankfull width, bankfull height above the water surface and thalweg depth. Value: mean bankfull width divided by mean bankfull depth (bankfull height plus thalweg depth). | Entered | 7.5 | |
| High flow dynamics | Impervious surface (%) | D | Percent of the delineated watershed in impervious surface, computed by DEEP. No field measurement. | Available | 3.0 | NLCD 2021 |
| Floodplain connectivity | Bankfull width:depth ratio | F | At each transect, measure bankfull width, bankfull height above the water surface and thalweg depth. Value: mean bankfull width divided by mean bankfull depth (bankfull height plus thalweg depth). | Entered | 7.5 | |
| Floodplain connectivity | Bankfull height above channel (m) | F | At each transect, measure the bankfull height above the present water surface in m, with a rod at the water's edge and a clinometer held level at bankfull. Value: mean of the 11 heights. | *Measure in the field* | | |

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| Function | Metric | F/D | How it is measured | Status | Value | Source |
|---|---|---|---|---|---|---|
| Hyporheic connectivity | Sand + fines (%) | F | Record the particle size class at 5 evenly spaced points on each of the 11 transects and the 10 cross-sections midway between them (105 particles). Value: percent sand or finer. | *Measure in the field* | | |
| Hyporheic connectivity | Embeddedness (%) | F | At 5 evenly spaced points on each of the 11 transects, estimate the percent of the particle buried in sand or finer sediment. Value: mean of the 55 estimates. | *Not applicable* | | |
| Channel evolution | Bankfull width:depth ratio | F | At each transect, measure bankfull width, bankfull height above the water surface and thalweg depth. Value: mean bankfull width divided by mean bankfull depth (bankfull height plus thalweg depth). | Entered | 7.5 | |
| Channel evolution | Bankfull height above channel (m) | F | At each transect, measure the bankfull height above the present water surface in m, with a rod at the water's edge and a clinometer held level at bankfull. Value: mean of the 11 heights. | *Measure in the field* | | |
| Channel and floodplain dynamics | Sinuosity | F | Take a compass backsight bearing and the distance between each pair of neighboring transects. Value: channel length along those segments divided by the straight-line distance between the reach ends. | *Measure in the field* | | |
| Channel and floodplain dynamics | Bank angle (degrees) | F | On both banks of each transect, lay a pole about 1 m long against the bank and read its angle from horizontal with a clinometer, in degrees. Value: mean of the 22 readings. | Entered | 12.5 | |
| Sediment continuity | Relative bed stability (log) | F | Computed: the base 10 log of the geometric mean particle diameter divided by the largest diameter the bankfull flow can move, from the particle count, reach slope, bankfull depth, wood tally and residual pools (NRSA physical habitat calculations). | *Measure in the field* | | |
| Sediment continuity | Sand + fines (%) | F | Record the particle size class at 5 evenly spaced points on each of the 11 transects and the 10 cross-sections midway between them (105 particles). Value: percent sand or finer. | *Measure in the field* | | |
| Bed composition and bedform dynamics | Substrate size (log mm) | F | From the 105-particle size class count (as for sand and fines). Value: base 10 log of the geometric mean particle diameter in mm, from the size classes (NRSA physical habitat calculations). | *Measure in the field* | | |
| Bed composition and bedform dynamics | Large wood volume (m3/100m) | F | Between each pair of transects, tally every piece of wood at least 10 cm in diameter and 1.5 m long in or above the bankfull channel, by length and diameter class. Value: wood volume per 100 m of channel, from the tally. | *Measure in the field* | | |
| Light & thermal regime | Chlorophyll a (ug/L) | F | Water-column chlorophyll a (ug/L) from a 2 L mid-channel grab sample at the X-site, collected in an amber bottle, filtered and analyzed in a lab. | *Measure in the field* | | |
| Light & thermal regime | Canopy density, mid-channel (%) | F | At mid-channel on each transect, read a convex spherical densiometer facing upstream, downstream, left and right (covered points of 17). Value: mean of the 44 readings, as a percent. | Entered | 17.5 | |
| Carbon processing | Riparian veg cover, woody (frac) | F | In a 10 m by 10 m plot on each bank at each transect, estimate the areal cover class of woody vegetation in the canopy, understory and ground layer. Value: summed woody cover of the 3 layers, averaged over the 22 plots, as a fraction. | *Measure in the field* | | |

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| Function | Metric | F/D | How it is measured | Status | Value | Source |
|---|---|---|---|---|---|---|
| Nutrient cycling | Total nitrogen (mg N/L) | F | Total nitrogen (mg N/L), lab analysis of the unfiltered mid-channel grab sample from the X-site, kept on ice. | *Measure in the field* | | |
| Nutrient cycling | Total phosphorus (ug/L) | F | Total phosphorus (ug/L), lab analysis of the mid-channel grab sample from the X-site, kept on ice. | *Measure in the field* | | |
| Water & soil quality | Specific conductivity (uS/cm) | F | Specific conductance (uS/cm). NRSA reports the lab value of a mid-channel grab sample at the X-site (the reach center); a calibrated multi-parameter meter reads the same quantity in the stream there. | *Measure in the field* | | |
| Water & soil quality | Turbidity (NTU) | F | Turbidity (NTU), lab analysis of the mid-channel grab sample from the X-site. Sample before anyone walks the channel upstream. | Entered | 22.5 | |
| Habitat provision | Fast-water / riffle habitat (%) | F | At each thalweg station, record the channel unit (falls, cascade, rapid, riffle, glide, pool, dry). Value: percent of stations in falls, cascade, rapid or riffle. | *Measure in the field* | | |
| Habitat provision | Natural fish cover (frac) | F | In the 10 m of channel centered on each transect, estimate the areal cover class of each fish cover type on the form (wood, brush, overhanging vegetation, undercut banks, boulders, others). Value: summed cover of the natural types, averaged over the transects. | *Not applicable* | | |
| Population support | Benthic total taxa richness | F | From the 11-transect composite D-frame sample (500 um mesh) and its lab subsample. Value: total number of taxa. | *Measure in the field* | | |
| Community dynamics | Shannon diversity (H') (H') | F | From the 11-transect composite D-frame sample (500 um mesh) and its lab subsample. Value: Shannon diversity of the taxa counted. | *Measure in the field* | | |
| Community dynamics | Tolerant individuals (%) ~ HBI | F | From the 11-transect composite D-frame sample (500 um mesh) and its lab subsample. Value: percent of individuals in pollution-tolerant taxa. | Entered | 27.5 | |
| Watershed connectivity | Mapped dams within one mile (dams) | D | Dams in the USACE National Inventory of Dams within 1 mile of the site, counted by DEEP. Note in the field any barrier the inventory does not show. | Available | 3.0 | NLCD 2021 |
| Population support | Native fish taxa richness | | | *Not scored* | | Insufficient reference support |
| Low flow and baseflow dynamics | Residual pool depth (cm) | | | *Not scored* | | Insufficient reference support |

F is measured in the field and D is answered from the desk. Available: the value was pulled and is printed on the field worksheet. Unavailable: no source answered, so measure or estimate it on site. Reference only: the value is shown and kept out of the score, because the curve was fitted on another data source. Not scored: the assessment withholds the metric for insufficient reference support.